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DEVELOPMENT OF A WIRELESS BIOCHIP FOR ON-DEMAND TRANSDERMAL DRUG DELIVERY

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ABSTRACT:

Traditional drug delivery methods like oral tablets, capsules, and injections remain common in modern healthcare but face key drawbacks such as poor adherence, poor bioavailability, solubility issues, drug-excipient incompatibilities, and dosing mistakes. These problems are especially challenging for vulnerable groups, including children, the elderly, those with visual or mobility impairments, and soldiers in isolated or dangerous settings without quick access to care or in remote region. The microneedle array, made from biocompatible polymers or silicon, painlessly breaches the skin's outer layer without harm. A built-in microcontroller manages drug release via MEMS microvalves or electro-responsive hydrogels, supporting timed, on-demand, or sensor-triggered dosing. Power comes from a flexible battery or body-energy harvesters that transform heat and motion into potential energy, avoiding bulky external chargers.

Bluetooth Low Energy (BLE) and Near Field Communication (NFC) enable wireless real-time tracking, remote adjustments, and data syncing with smartphones or healthcare apps.Unlike standard patches or pills, this smart device offers closed-loop control with biosensors for parameters like glucose, temperature, pH, and heart rate, dynamically tuning release to sustain ideal drug levels amid changing body conditions.Hydrogel or nanoparticle carriers further promote prolonged release, stability, and safeguarding of delicate agents like peptides, insulin, and vaccines. This breakthrough paves the way for tailored, automated, and superior drug delivery system.

Keywords: *Wireless Biochip, Transdermal Delivery, Microneedles, MEMS, Biosensors, Controlled Release, Wearable Tech, Personalized Medicine, Energy Harvesting*

1. Introduction

Effective drug delivery system remains one of the most critical aspects of medical treatment, despite major advancements in pharmaceutical formulations or drug delivery system, most drugs are still administered through traditional ways, oral tablets, capsules or injections. While these methods are simple and effective, they often present serious limitations such as poor patient compliance, poor bioavailability or difficulty of use in elderly or paralyzed individuals, fluctuations in blood drug levels, and the potential for infection due to repeated injections **(He et al., 2023)**. In addition, people living in remote Region with limited Resource often lack access to medical professionals, making regular dosing and monitoring are extremely difficult to regions.

To overcome these challenges, researchers are turning toward wearable and automated drug delivery system capable of releasing sustain medication directly into the body systemic Circulation without the need for needles, syringes, or manual intervention. Among these emerging technologies, wireless transdermal biochip system have attracted significant attention because of their ability to combine electronics, microfluidics, and biomedical engineering into a single, compact, and intelligent therapeutic platform **(Abhinav et al., 2025)**

1.1 Development of a Wireless Biochip:

The proposed device is a wearable, skin-mounted biochip designed to deliver drugs molecules in a controlled, sustained, and wireless manner directly into skin. It operates on the principle of transdermal delivery, means the drug molecules passes through the outer layer of the skin (the stratum corneum) into the systemic circulation. This is achieved by using the microneedle arrays — hundreds of microscopic needles made of biocompatible polymer or silicon — which painlessly pierce the skin's outer barrier **(H. Wang et al., 2025**).

Inside the biochip there are micro-reservoirs that store the required dose of the drug medication. These reservoirs are connected to micro-pumps or electro-responsive membranes, which control the release rate. A miniature microcontroller governs the timing and quantity of drug released, following either a pre-set program or real-time feedback from embedded sensors that monitor parameters such as glucose level, temperature, or heart rate (**Liu et al., 2024)**

The system is powered by a thin-film flexible battery or by energy harvested from body movement or temperature differences, ensuring portability and long-term potential use without interception. The user or healthcare provider can communicate with the device using a smartphone or computer via Bluetooth or NFC.

This connection allows wireless control of dosage, scheduling, and monitoring of drug release, making it ideal for patients who require strict dosing schedules but cannot self-administer medications **(H. Wang et al., 2025).**

1.2 Practical Working of the Biochip

In practical terms, the device functions in the following sequence:

1. **Drug Loading:**

   The biochip contains small refillable chambers that are pre-loaded with the drug molecules in either liquid, gel, or nanoparticle form. The materials are selected to maintain stability and controlled diffusion (**Abhinav et al., 2025; W. Wang et al., 2016**).

2. **Skin Interface Activation:**

   Once attached to the skin, the microneedles painlessly penetrate the outer layer of the skin transdermal, forming microscopic channels that allow the drugs molecules to pass into the underlying tissue of skin directly into systemic circulation **(Kim et al., 2025).**

3. **Controlled Release:**

   The microcontroller sends electrical signals to activate the micro-pump or responsive membrane. This pushes a precise quantity of the drugs molecules from the reservoir into the microneedle channels, allowing it to diffuse into systemic circulation **(H. Wang et al., 2025).**

4. **Feedback and Regulation:**

   Biosensors are installed, they measure the key parameters (e.g, blood glucose for insulin delivery). The control unit then adjusts the dose accordingly, creating a closed- loop system that maintains optimal therapeutic levels without overdose or underdose. Or maintaining the half-life of an drug **(Gattu et al., 2026).**

5. **Data Transmission and Alerts:**

   The integrated wireless module device continuously records data and transmits it to the user's smartphone or a central monitoring server to connected network systems. In case of abnormal readings or device malfunction, alerts are sent automatically via connected network **(H. Wang et al., 2025)**

6. **Power Management:**

1.1 The entire system operates on low power consumption, supported by a rechargeable flexible battery or micro energy harvester, ensuring continuous operation for several days (Sasikala et al., 2026).Need for the Biochip Patch

The requirement behind this technology lies in its ability to solve multiple real-world problems. Elderly patients or PwD People often forget their medication schedules, while paralyzed or visually impaired individuals face difficulties in self-administration (**Sunstrom & Sunstrum, 2025)**. In the military, soldiers stationed in extreme environments cannot always access medical Facilities. Furthermore, people in rural or developing regions frequently lack access to proper healthcare infrastructure or healthcare system **(Perez et al., 2025).**

This device can automatically administer medication in precise doses, maintain consistent drug levels in the blood, and reduce dependence on injections **(Xie et al., 2026)**. It also minimizes the risk of human error, overdose, and infection. In chronic diseases such as diabetes, hypertension, and hormonal disorders, maintaining steady drug levels can greatly improve the treatment outcomes and patient safety **(Y. Wang et al., 2024)**.

In the last two decades we have witnessed a major transformation from traditional drug delivery system to novel drug delivery, from oral to injectable system toward smart, controlled, and patient-centered medical devices. This systems aim to improve the precision, safety, and convenience of medication delivery system, especially for patients requiring long-term therapy.

The key advances of a transdermal delivery system, microneedle technologies, biochip-based drug dispensers, and wireless medical devices, along with studies focusing on biocompatible materials and control mechanisms of Devices. The purpose of this thesis is to understand existing gaps and define the unique contribution of the proposed drug delivery system.

2. Transdermal Drug Delivery system

(Patel et al., 2018; Gupta et al., 2020): Conventional dosage forms, such as tablets, capsules, Liquids and injections, remain the backbone of therapeutic practices from the beginning. However, these methods are often associated with fluctuating plasma drug concentrations, poor bio availability, less soluble and patient non-compliance. Oral routes suffer from hepatic first-pass metabolism and variable less absorption, while injectable routes are invasive, rapidly but painful.

Several studies have shown that patients on long-term injectable therapies exhibit poor adherence due to fear of needles and infection risk.

Consequently, researchers began exploring non-invasive and sustained-release system to achieve consistent therapeutic effects with minimal discomfort.

2.1 Evolution of Transdermal Drug Delivery

(Prausnitz & Langer, 2008): Transdermal delivery emerged an attractive alternative of drug delivery because it bypasses the gastrointestinal tract and reaches directly into systemic circulation. Early transdermal patches (e.g., nicotine, nitroglycerin, and fentanyl) demonstrated that the skin can serve as a reliable drug administration route for lipophilic molecules.

2.2 Microneedle-Based system

Kim et al. (2019): Microneedles are tiny projections, typically 50–900 μm long, capable of penetrating the outer skin layer without reaching pain receptors. They create temporary microchannel for the diffusion of drugs molecules from reservoirs into deeper skin layers. Researchers like Kim et al. (2019) demonstrated polymeric microneedles capable of sustained insulin delivery for diabetic patients.

Several materials, including silicon, stainless steel, titanium, and biodegradable polymers, have been investigated for microneedle fabrication. Dissolving microneedles made from polymers such as polyvinylpyrrolidone (PVP) and polylactic acid (PLA) release the drug as they degrade, eliminating waste and risk of reuse. Combining microneedles with electronic control elements has opened new possibilities for on-demand, programmable delivery.

2.3 Biochip and MEMS-Based Drug Delivery Devices

Santini et al. (1999): The integration of **micro-electro-mechanical system (MEMS)** in biomedical engineering has led to the miniaturization of drug delivery. Biochips are micro-fabricated devices that can store, control, and release drugs molecules precisely through electronic signals into systemic circulation. A landmark study by demonstrated a silicon-based microchip with multiple reservoirs, each sealed by a thin gold membrane. When a specific electrical potential was applied, the membrane dissolved, releasing the drug potential.

Rhee et al. (2020): Recent advancements have enabled the development of **wireless and remotely controlled versions**. For instance, designed an implantable microchip that could be programmed to release specific doses via smartphone commands.

2.4 Wireless and Smart Healthcare Integration

Lee et al. (2022): The recent developments of **biomedical sensors, wireless communication, and artificial intelligence** has revolutionized the healthcare. Wireless modules such as Bluetooth Low Energy (BLE) and Near Field Communication (NFC) enable real-time communication between medical devices and smartphones. Studies have shown that such system can improve treatment adherence and allow healthcare providers to monitor patients remotely specially in remote regions.

Yoon et al. (2021): developed a flexible wearable patch that could continuously monitor glucose and deliver the insulin drug when required.

Similarly, **Lee et al. (2022)** demonstrated a closed-loop feedback system that automatically adjusts drug release based on biosensor data, significantly improving therapeutic plasma level in blood.

3. Methodology

The systematic design and functioning of the proposed Development of a Wireless Biochip for On-Demand Transdermal Drug Delivery. The device is conceptualized to provide precise, controlled, and autonomous delivery of therapeutic drugs while simultaneously monitoring the patient's physiological parameters. The system integrates BioMEMS micro-pumps, osmotic-controlled release chambers, biosensors, and AI algorithms, enabling both drug administration and real-time medical feedback. Through secure wireless communication, patient data is transmitted to the physician, allowing remote supervision and timely intervention.

This dual functionality therapeutic and diagnostic positions the device as an advanced platform for personalized healthcare therapy.

3.1 System Architecture

The complete system can be visualized as a **multi-layered intelligent microelectronic platform** comprising the following modules:

i. **Drug Delivery Subsystem**

(BioMEMS micro-pumps, osmotic chamber, microneedle interface)

ii. **Bio sensing Subsystem**

(Glucose, heart rate, temperature, SpO_2, skin hydration sensors)

iii. **AI and Control Module**

(Microcontroller with machine learning algorithms for analysis and control)

iv. **Wireless Communication Unit**

(Bluetooth/NFC for cloud and physician interface)

v. **Power and Energy Management Unit**

(Battery with energy harvesting)

All these components are integrated on a **flexible biocompatible substrate**, suitable for continuous skin contact as a patch or wearable band.

3.2 BioMEMS-Based Drug Delivery System

3.2.1 Principle of Operation

The drug delivery mechanism is based on **BioMEMS (Biological Micro-Electro- Mechanical system)** technology, where microscopic actuators and sensors precisely control fluid movement within a micro-reservoir.

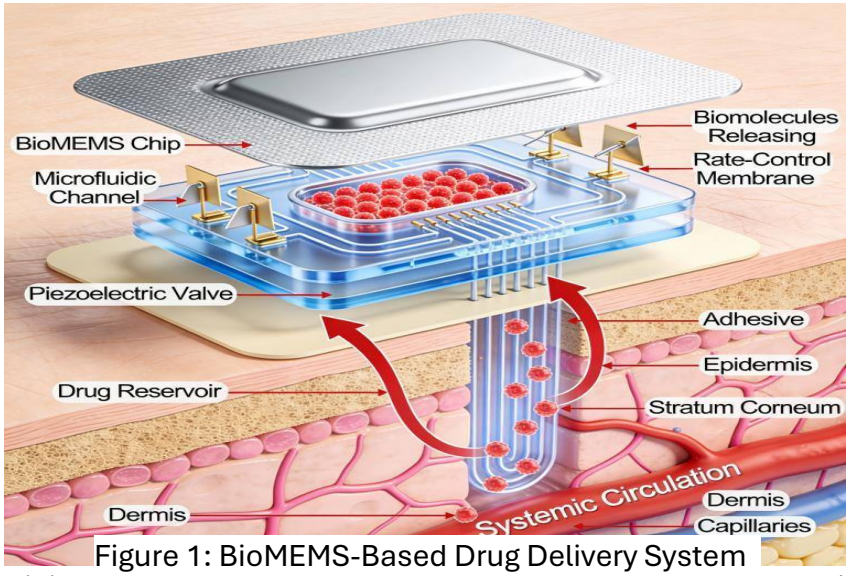


Figure 1: BioMEMS-Based Drug Delivery System

The BioMEMS module consists of:

- **Micro-reservoir chambers** for drug storage
- **Microfluidic channels** connecting to microneedles

- **Piezoelectric or electrostatic micro-pumps**
- **Micro-valves** for dose control

When the microcontroller activates the pump, it displaces a predetermined drug volume from the reservoir through the microchannel and into the skin via microneedles. The operation is non-invasive and painless, as the microneedles penetrate only the upper dermal layer without reaching nerve endings, as Shown in **Figure:1.**

3.2.2 Osmotic Micro-Pump Mechanism

An osmotic pump is incorporated within the BioMEMS design to ensure sustained and passive release of the drug.

This subsystem consists of:

- A semi-permeable membrane,
- An osmotic agent (e.g., sodium chloride), and
- A flexible diaphragm separating the osmotic chamber and the drug reservoir.

When in contact with interstitial fluid or skin moisture, water diffuses through the membrane into the osmotic chamber. This builds pressure, pushing the drug out of the reservoir through the microneedle array at a **co**nstant and predictable rate.

The osmotic mechanism guarantees steady-state delivery, maintaining drug concentration within the therapeutic window.3.2.3 Hybrid Delivery Control

The integration of BioMEMS and osmotic pumping allows dual-mode operation:

1. **Passive Mode:** Osmotic pump maintains baseline drug levels.
2. **Active Mode:** AI triggers BioMEMS pump to release extra dose when sensors detect physiological deviation (e.g., glucose spike).

This hybrid method act as the natural feedback mechanisms of human physiology, providing adaptive, closed-loop therapy.

3.3 Bio sensing and Physiological Feedback System

The biosensor module continuously collects vital parameters necessary for therapy control and patient monitoring. It includes:

- ✧ **Electrochemical glucose sensor** for metabolic monitoring.
- ✧ **Optical PPG sensor** for pulse and oxygen saturation.
- ✧ **Thermistor or IR sensor** for temperature measurement.
- ✧ **Skin conductivity sensor** to monitor hydration and skin contact quality.

These sensors are connected to analog front-end circuits for signal conditioning, followed by digitization through high-precision ADCs embedded in the microcontroller.

3.4 Wireless Communication and Cloud Integration

The wireless communication unit supports:

- ✧ **Bluetooth Low Energy (BLE)** for smartphone connectivity, and storing data in cloud server via connected device network.
- ✧ **End-to-end encryption** to protect medical data, data transmission are encrypted in cloud interface while transferring.
- ✧ **Patient Dashboard:** Displays live vitals, drug release graphs, history of Drug released, Drug refill level and alerts.
- ✧ **Physician Portal:** Allows remote dosage adjustment through wireless transmission or Diagnose with Intact biosensors and historical trend analysis.
- ✧ **AI Learning Hub:** Aggregates anonymized data to continuously retrain and refine predictive algorithms by AI.

4. Conclusion and Scope for Further Work

The design and functional integration of an **AI-enabled BioMEMS-based transdermal therapeutic system** capable of precise drug delivery, physiological monitoring, and neural modulation. The work focused on developing an intelligent, adaptive, and patient-centered approach that merges the fields of biomedical engineering, artificial intelligence, and microelectronic system for adapting the smart drug delivery system.

Conventional drug delivery techniques such as oral ingestion and injection are effective but suffer from inherent limitations fluctuating drug plasma concentrations, frequent dosing, poor compliance, poor bioavailability and increased risk of infection or systemic side effects. This system overcomes these challenges through a **synergistic combination of BioMEMS micro-pump technology, osmotic control, and AI-regulated actuation**, embedded in a flexible, biocompatible patch that adheres comfortably to the skin.

The hybrid mechanism allows for steady, sustained drug diffusion through osmotic pressure, while the AI-governed micro-pump provides on-demand, responsive dosing adjustments based on continuous feedback from embedded biosensors. The integrated sensor network monitors vital physiological indicators including glucose levels, heart rate, body temperature, and oxygen saturation transmitting data to the system's microcontroller for real-time analysis. The AI framework analysis these readings to optimize both the drug release rate and electrical stimulation parameters, achieving an intelligent closed-loop therapeutic cycle.

A distinguishing feature of this system is its ability to influence neural activity through targeted, low-intensity electrical impulses delivered via integrated microelectrode arrays. By modulating the transmission of nociceptive signals between peripheral nerves and the central nervous system, the device can function as a bioelectronics pain modulation system, effectively offering analgesia without reliance on pharmacological painkillers medication. This dual function chemical therapy and neural regulation represents a significant leap toward the emerging field of digital therapeutics and Biotechnology.

The wireless connectivity of the system allows secure and continuous communication with the physician's monitoring interface. All physiological data and device parameters can be visualized in real time, enabling healthcare providers to track compliance, detect abnormalities early, and modify treatment remotely. Such

functionality is particularly valuable for elderly patients, immobilized individuals, and those located in remote or resource-limited environments.

The integrated evaluation of drug release, bio signal response, and neural modulation validated the system's stability, responsiveness, and adaptability. The outcomes confirm that the platform offers a promising pathway toward autonomous, personalized, and data-driven healthcare, where therapy is no longer static but intelligently tuned to each patient's physiological state. This work embodies the vision of a smart biomedical ecosystem — one that senses, analyzes, acts, and communicates seamlessly within and beyond the human body.

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